

Bill Charge-Line Decomposition Spec (Phase 1)

PREPARED BY Energy Hippo implementation team

How every Northlake monthly bill decomposes into `bill_charge_line` rows (EEMSuite `BillItem`), so cost reports (`rp_CostBreakOut`, `rp_BillingCharges`, `rp_RateComponentMap`) and GL coding (Phase 3) have itemized data, and every bill's lines provably reconcile to its total.

Scope boundary. Phase 1 fixes the **line structure + reconciliation rules** (below). Phase 2 (rate engine) supplies the **determinant rates** that make each line computable from usage; until then, lines are reconciled against the known bill total. The per-provider charge **codes** already live in each `data/providers/<id>.yaml` `billCharges` block — this spec maps them to the `bill_charge_line.charge_type` enum, the EEMSuite determinant kind, and the decomposition arithmetic.

Not yet decomposed here: the Central Plant's VED-TOU-PRI electric, VED-AL-1 unmetered lighting, SGU-GL-1 large-volume gas, RCU-IRR irrigation and RCU-FIRE fire-service schedules.

charge_type → EEMSuite determinant kind

`bill_charge_line.charge_type` (from `implementation/synthetic-source-model.pdf`) maps to the rate-engine determinant math kind (see [subsystem-gap-analysis.pdf](#) §3):

charge_type	EEMSuite determinant kind	Sign	Notes
fixed / customer	400 Fixed by period (or 300 Fixed by day)	+	meter/facility/customer/base charge
energy	200 Charge period (energy)	+	flat \$/unit consumption
tou	200 with a TOU schedule	+	one line per TOU period
seasonal	200 with a season schedule	+	rate chosen by month
tier	200 with multiple blocks	+	one line per block consumed
demand	200 with <code>IsDemand=1</code>	+	billed on peak (max-interval)
rider / surcharge	700 Bill adj (%) or 200 (per-unit)	+	program/public-purpose riders
tax	800 Taxes (% on subtotal)	+	applied after all ≤800 charges
credit	200 with negative block (or 700 neg %)	−	net-metering / generation credit
adjustment	200 (precomputed) or post-script	±	e.g. power-factor adjustment (synthetic stand-in — engine has no PF math)
allocation	200 / 400 (internal)	+	internal cost-recovery line (district energy)
generation	200 on a generation channel	+	PPA energy priced on kWh produced
minimum / maximum	500 / 600	±	floor/cap (introduced in Phase 2)
informational	none (not a money line)	0	e.g. REC count; excluded from the bill total

Per-provider decomposition

Each row = one `bill_charge_line` on every monthly bill for that provider/rate. `qty source` is what the line's quantity comes from; `rate source` points at the `data/providers/<id>.yaml` rate component.

Valley Electric District — VED-TOU-GS (electric)

line	charge_type	commodity	qty source	rate source
energy_peak	tou	electric	peak kWh	components.energy.peak
energy_part_peak	tou	electric	part-peak kWh	components.energy.part_peak
energy_off_peak	tou	electric	off-peak kWh	components.energy.off_peak
demand_charge	demand	electric	max 15-min kW	components.demand
facility_charge	fixed	electric	1 / month	components.facilityCharge
power_factor_adjustment	adjustment	electric	precomputed	(needs rate param — Phase 2; synthetic stand-in)
public_purpose_program	rider	electric	% of energy or flat	(needs rate param — Phase 2)
utility_users_tax	tax	electric	% of pre-tax subtotal	(needs rate param — Phase 2)

- **TOU split for interval-metered buildings** (Science Center, Student Center, Lakeview Residence Hall, Recreation and Aquatics Center): peak/part/off kWh come from the 15-min interval data classified by the TOU calendar (summer has peak/part/off; winter has part/off only — see [seasons](#)).
- **TOU split for monthly-only buildings** (Admin Hall, Library, Parking Structure, Cedar Row A, Cedar Row B, Cedar Row Common House): no interval data, so the bill carries TOU-split kWh from **per-TOU meter registers** (three register reads), not from interval data. (This is how real TOU meters bill non-interval accounts.) Winter bills omit the peak line.
- FY2025 bills (period ≥ 2025-01) use VED-TOU-GS-FY25 rates (the `provider_rate_change` scenario); line structure is identical.

Sierra Gas Utility — SGU-GN-1 (natural gas)

line	charge_type	commodity	qty source	rate source
customer_charge	fixed	natural_gas	1 / month	components.customerCharge
gas_commodity	seasonal	natural_gas	therms	components.commodity.winter/summer
gas_transport	energy	natural_gas	therms	components.transport
public_purpose_surcharge	rider	natural_gas	% or flat	(Phase 2)
utility_users_tax	tax	natural_gas	% of pre-tax subtotal	(Phase 2)

- **Seasonal selection:** the `gas_commodity` rate is winter (Nov–Mar) or summer (Apr–Oct) by the bill's period month.

- **Estimated → corrected scenario:** the estimated bill carries the same line structure with estimated therms; the corrected actual bill supersedes it with the same lines at actual quantities.
- `ccf` → therms uses `thermFactor 1.027` (the `gas_commodity/gas_transport` quantity is therms).

River City Utilities — RCU-W-TIER + RCU-S-VOL + RCU-SW-ERU (water/sewer/stormwater)

One account, one statement, three commodities:

line	charge_type	commodity	qty source	rate source
water_meter_charge	fixed	water	meter size (1/2/4 in)	RCU-W-TIER.meterCharge
water_tier1	tier	water	kgal in 0–50	RCU-W-TIER.tiers[0]
water_tier2	tier	water	kgal in 50–200	RCU-W-TIER.tiers[1]
water_tier3	tier	water	kgal over 200	RCU-W-TIER.tiers[2]
sewer_base	fixed	sewer	1 / month	RCU-S-VOL.baseCharge
sewer_volume	energy	sewer	metered water kgal	RCU-S-VOL.volume
stormwater_eru	fixed	stormwater	ERU count	RCU-SW-ERU.eruCharge

- **Tier blocks:** the three `water_tier*` lines partition the period's water kgal (each clamped to its block) and together equal total metered water. Buildings with no usage in a tier omit that line.
- **Derived sewer:** `sewer_volume` quantity = metered water kgal (apartments capped at a winter average per the provider note). Cedar Row accounts have no stormwater line.
- **Duplicate-import scenario:** re-importing the combined file must not duplicate any of these lines.

Northlake Thermal Plant — NTP-ALLOC-CHW + NTP-ALLOC-STEAM (internal allocation)

Cost-center "bills" (`account_kind = internal_cost_center`), not utility bills:

line	charge_type	commodity	qty source	rate source
chilled_water_consumption	allocation	chilled_water	ton-hours	NTP-ALLOC-CHW.consumption
chilled_water_demand	demand	chilled_water	peak tons	NTP-ALLOC-CHW.demand
steam_consumption	allocation	steam	klb	NTP-ALLOC-STEAM.consumption
fixed_oandm_allocation	allocation	(plant)	fixed monthly share	(annual plant O&M recovery; Phase 2 rate param)

- Science Center and the Recreation and Aquatics Center have both chilled-water lines (each has a ton-hour/tons BTU meter); Admin Hall, Library, Student Center and Lakeview Residence Hall carry `steam_consumption` only.
- These behave like bills for rollups (`baseline_monthly_bills` secondary).

Helios Onsite Solar — HOS-PPA-2024 (PPA invoice)

line	charge_type	commodity	qty source	rate source	sign
ppa_energy	generation	solar_generation	generated kWh	components.ppaEnergy (escalating)	+
net_metering_credit	credit	solar_generation	exported kWh	components.netMeteringCredit	-
rec_transfer	informational	solar_generation	REC count	recOwnership	0

- **Escalator:** ppaEnergy rate rises 2.0%/year (annualEscalator) — 2025 bills use the escalated rate.
- **Net invoice:** total = ppa_energy - net_metering_credit. The rec_transfer line carries a REC count, **not money**, and is excluded from the total (informational).
- **Inverter-outage backfill:** generation gaps reduce ppa_energy qty until backfilled.

Decomposition rules (cross-cutting)

1. **Reconciliation:** $\sum(\text{bill_charge_line.amount where charge_type} \neq \text{informational}) = \text{bill.total_cost}$, exactly, per bill.
2. **Quantity reconciliation:** per commodity, the consumption lines' quantities reconcile to the bill's usage_quantity (TOU lines sum to total kWh; tier lines sum to total kgal).
3. **Ordering:** taxes (tax, kind 800) apply to the sum of all preceding non-tax lines; if multiple taxes exist they compound in line order.
4. **Signs:** credit lines are negative; informational lines are zero.
5. **Line presence is conditional:** omit a line whose quantity is zero (e.g. a winter VED peak line, an empty water tier, a Cedar Row stormwater line).
6. **Estimated/corrected/duplicate** scenarios preserve the same line structure; corrections supersede, duplicates must not double lines.

Worked examples (illustrative — exact rates finalized in Phase 2)

VED Science Center, a summer month (interval-metered):

line	qty	rate	amount
energy_peak	12,000 kWh	0.241	2,892.00
energy_part_peak	18,000 kWh	0.168	3,024.00
energy_off_peak	30,000 kWh	0.112	3,360.00
demand_charge	220 kW	18.40	4,048.00
facility_charge	1	64.00	64.00
power_factor_adjustment	—	—	85.00
public_purpose_program	—	—	120.00
utility_users_tax	5.0% × 13,593.00	—	679.65
total			14,272.65

energy lines sum to 60,000 kWh (= bill usage); lines sum to total. ✓

RCU Admin Hall, 2-inch meter, 120 kgal water:

line	qty	rate	amount
water_meter_charge	2in	96.00	96.00
water_tier1	50 kgal	4.85	242.50
water_tier2	70 kgal	6.40	448.00
water_tier3	0 kgal	8.15	(omitted)
sewer_base	1	28.50	28.50
sewer_volume	120 kgal	7.95	954.00
stormwater_eru	18 eru	11.20	201.60
total			1,970.60

water tiers sum to 120 kgal (= metered water = sewer volume basis). ✓

expectedAssertions

- `charge_lines_sum_to_total` — per bill, non-informational lines sum to `total_cost` (tolerance: 1 cent).
- `usage_lines_reconcile` — per commodity, consumption-line quantities sum to the bill's usage quantity.
- `tier_blocks_partition_usage` — RCU water tier lines partition total water kgal with correct block bounds (50 / 200).
- `tou_lines_sum_to_kwh` — VED TOU lines sum to total kWh; winter bills carry no peak line.
- `credit_is_negative` — HOS `net_metering_credit` amount < 0; `rec_transfer` amount = 0.
- `tax_on_subtotal` — `utility_users_tax` = `tax_rate` × (sum of preceding lines).